

Compressive strength and damage mechanisms of 2D-C/SiC composites at high temperatures

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ABSTRACT

Compressive strength of 2D-C/SiC composite was investigated from room temperature (RT) to 1600 °C at present work. Damage evolution was investigated by conducting loading/unloading tests at RT and the damage mechanisms were elucidated by observing the fracture morphology. It is found that compressive strength of 2D-C/SiC was retained until 1200 °C and then decreased with increasing temperature. The variation of compressive strength is closely related to the degradation in matrix modulus. The compressive damage of 2D-C/SiC starts at the buckling of 0° fiber and is followed by opening and closing of original pores, initiation and growth of longitudinal interbundle cracks, separation of 90° fiber bundles by longitudinal cracks, matrix cracking from intrabundle pores, propagation of matrix cracks into 0° fiber bundles, connection of cracks in 0° fiber bundles and longitudinal cracks in 90° fiber bundles.

1. Introduction

Continuous carbon fiber reinforced silicon carbide (C/SiC) composites, an important structural material, are always used to manufacture thermal structural components, whose service temperature could reach as high as 1650 °C, because of their low density, excellent resistance to high temperature, and high strength retention [1–3]. The C/SiC components might be loaded in compression. Moreover, the C/SiC components will fail due to the compressive damage even when the components are loaded in tension [4,5]. For example, the C/SiC components containing the pin connections might be squeezed and fail in a compressive manner. Therefore, it is rather critical to investigate the compressive properties of C/SiC and further reveal its damage mechanisms.

Currently, extensive studies regarding the compressive strength and damage mechanisms have mainly focused on continuous-fiber-reinforced polymer matrix composites (PMCs). It is showed that the interface bonding and matrix compliance play the key roles in damage evolution [6–10]. A few studies found that similar phenomena were observed for C/SiC reinforced with a plain-woven carbon cloth (2D-C/SiC). However, the limited studies suggest there are significant differences between 2D-C/SiC and PMCs because of the intrinsic brittleness and much greater modulus of SiC matrix compared with the soft and tough matrix in PMCs. It is revealed by Wang et al. that the compressive damage of 2D-C/SiC lies in the cracking mechanisms [11], including ply delamination, separation of 0°/90° bundle, and cracking in 90° fiber

bundles. However, the previous studies did not reveal the variation in the compressive strength of 2D-C/SiC with increasing temperature and the corresponding damage mechanisms. Therefore, the aim of the present contribution is to reveal the variation in the compressive strength of 2D-C/SiC from room temperature (RT) to 1600 °C and to investigate the compressive damage mechanisms using a scanning electron microscope (SEM) and the loading/unloading tests at RT.

2. Experimental

2.1. Material preparation

2D-C/SiC was reinforced by laminated plain woven carbon cloths (1 K, T-300™, Toray industries Inc., Tokyo, Japan), as shown in Fig. 1(a). A pyrolytic carbon (PyC) interface with a thickness of 0.2 μm was deposited on the carbon fibers by chemical vapor deposition at 960 °C. SiC matrix was infiltrated into the preforms through chemical vapor infiltration (CVI). The volume fraction of fibers was about 45%, and the density was about 2.05 g/cm³. The details of the preparation of specimens can be found in Ref. [12]. Finally, the specimens, as shown in Fig. 1, were machined.

2.2. Compression tests

Compression experiments were performed at RT, 1200 °C, 1400 °C, and 1600 °C using a universal mechanical testing machine (RDL15,

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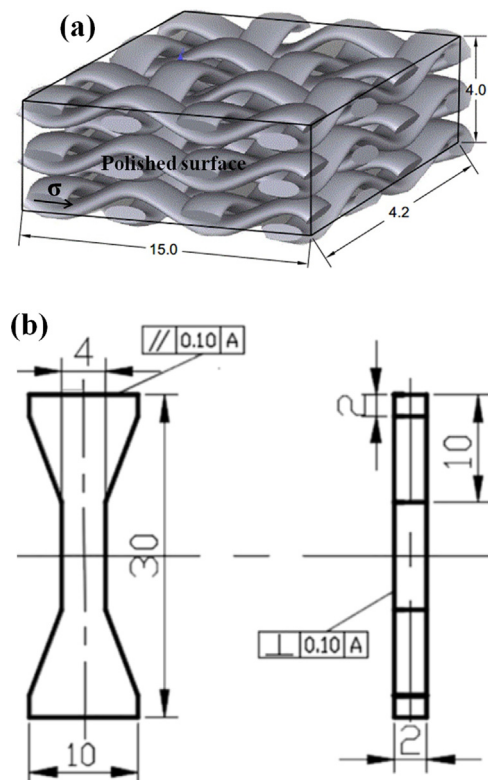


Fig. 1. Reinforcements and dimensions of 2D-C/SiC specimens. (a) plain woven clothes and dimensions of the loading/unloading specimens, (b) compressive specimens.

Changchun Institute of Mechanical Science, China). The loading speed was 0.5 mm/min. The compressive tests at RT were performed in air, and the compressive strength at high temperatures were measured in vacuum at a heating rate of 25 °C/min. Three specimens were tested for each testing condition. The fracture morphologies were observed using a SEM (S-4700, HITACHI, Tokyo, Japan). To observe the damage mechanisms, after the tests, a surface of some typical specimens displaying the laminated plain woven carbon cloths, as shown in Fig. 1(a), was ground using different SiC water sandpapers with different grits and polished using a diamond polishing paste with models W5, W3.5, W2.5, W1.5, W1, and W0.5 paste polishing suspensions.

2.3. Loading/unloading experiments

The compressive loading/unloading tests were carried out using a universal mechanical properties testing machine (Institute of Strength Problems, Academy of Sciences of Ukraine, CERA) with a loading rate of 0.5 mm/min at RT. Specimens with the shape of rectangular cross-section bar were used, as shown in Fig. 1(a). To clearly identify the damage within the composite, one side, shown in Fig. 1(a), was polished before the testing. First, the specimen was loaded to 100 MPa and unloaded. Then, the loading was increased to 200 MPa, and the specimen was unloaded. After each unloading, the specimen was observed using a SEM to determine the damage modes in the composite.

3. Results

3.1. Compressive strength at high temperatures

Fig. 2 shows the typical compressive stress–displacement curves of 2D-C/SiC at high temperature. Monotonic increase in stress with displacement was observed. When the stress reached the maximum, the curve linearly decreases, characteristics of typical brittle fracture.

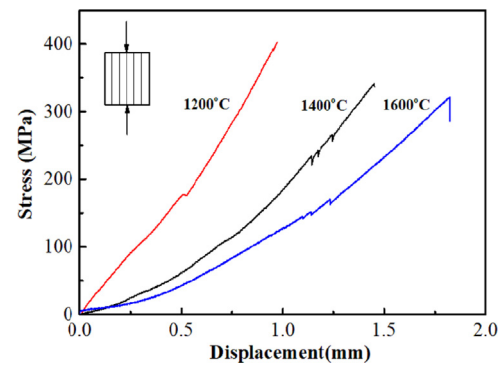


Fig. 2. Compressive stress–displacement curves of 2D-C/SiC.

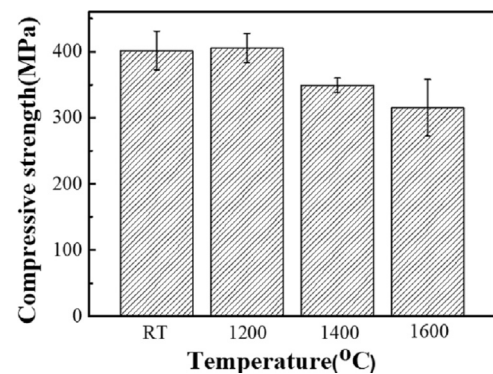


Fig. 3. Compressive strength of 2D-C/SiC composites at elevated temperature.

Fig. 3 shows the compressive strength of 2D-C/SiC composites at RT, 1200 °C, 1400 °C, and 1600 °C. The composite retains its compressive strength until 1200 °C. At a higher temperature, the compressive strength of the composite decreases with increasing temperature. This trend of compressive strength will be explained later.

3.2. Fracture morphology

Fig. 4 shows the typical morphology of 2D-C/SiC composite. A macroscopic crack inclined to the loading direction by $\sim 20^\circ$ can be observed after the failure of composite. Moreover, some cracks parallel

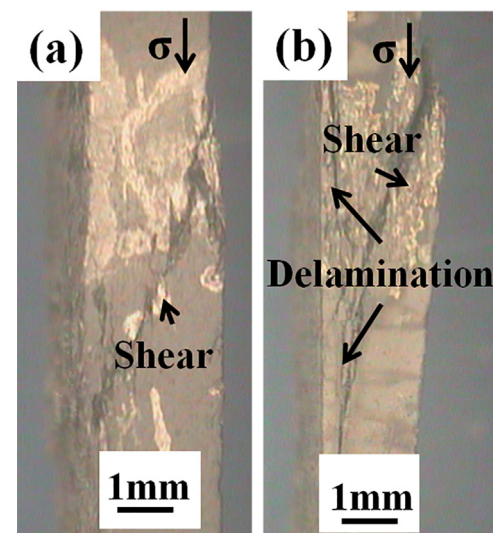


Fig. 4. Typical fractured specimens at high temperature (a) 1200 °C and (b) 1600 °C.

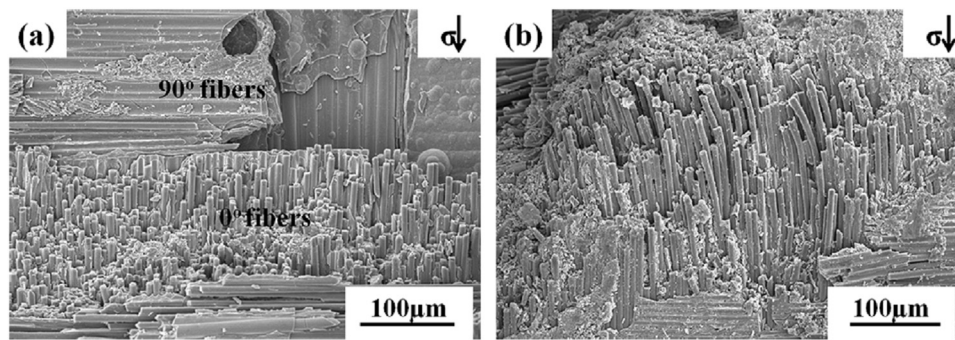


Fig. 5. Compressive fractography of 2D-C/SiC composites (a) 1200 °C and (b) 1600 °C.

to the loading direction were also observed, especially close to the specimen surfaces. The different cracks indicate that complex damage mechanisms operate in the composite. The inclined crack indicates that the composite fractured in the shear mode. However, the parallel cracks indicate that interlaminar debonding occurred between the different layers of woven reinforcement cloths.

Fig. 5 shows the microscopic morphology of fracture surface. Broken 0° fiber bundles and almost intact transverse bundles can be observed for the specimen tested at 1200 °C, as shown in Fig. 5(a). The broken 0° fiber bundles correspond to the areas failed in the shear manner, and the transverse bundles reflect the interlaminar debonding shown in Fig. 4. Moreover, the surface of the failed 0° fiber bundles is inclined in almost the same direction as the loading direction of the macroscopic crack shown in Fig. 4. This indicates that the fracture of composite is mainly contributed by 0° fiber bundles. At a higher temperature, the phenomenon is almost similar as that at 1200 °C. The difference lies on the broken 0° fibers with a longer pullout length and bent by the compressive load, as shown in Fig. 5(b), for the specimen that failed at 1600 °C.

To further reveal the compressive damage mechanisms, the surfaces, perpendicular to the fracture surface of some typical compression specimens, were polished and observed using a SEM. Fig. 6(a) shows the SEM micrograph of a specimen tested at 1200 °C. A main crack,

inclined to the loading direction, was clearly observed, as indicated by a dashed line. The 0° fiber bundles are completely fractured. Moreover, the 90° fiber bundles located at both sides of the main crack became loose. This indicates that many interbundle cracks are generated, accompanied by the growth of the main shear crack. When the samples were tested at higher temperatures, macroscopically, the specimens failed in a similar manner as that at 1200 °C. However, the microscopic feature of damage within the composite is different. Fig. 6(b) shows the cross-section of a specimen that failed at 1400 °C. The damage spread to a relatively wide zone. The phenomenon is completely different from that at 1200 °C, where the failure shows the main crack. The fracture zone is inclined to a similar angle as the loading direction. Within the fracture zone, the debonding of 0°/90° fiber bundles, interbundle cracks in 90° fiber bundles, and bending and kinking of 0° fiber bundles were observed. At 1600 °C, large open interbundle cracks and broken 0° fibers were observed. However, no matrix crack was observed around the broken fiber, as shown in Fig. 6(c). This indicates that a significant matrix flows during the compression. This should be related to the plastic deformation of SiC matrix at 1600 °C.

Further observations show that the compressive damage of composite is confined to the local area on the two sides of fracture surface. If the temperature is lower than 1400 °C, the damage of 0° fiber bundles develops as fiber bundle kinks and shear fracture, accompanied by

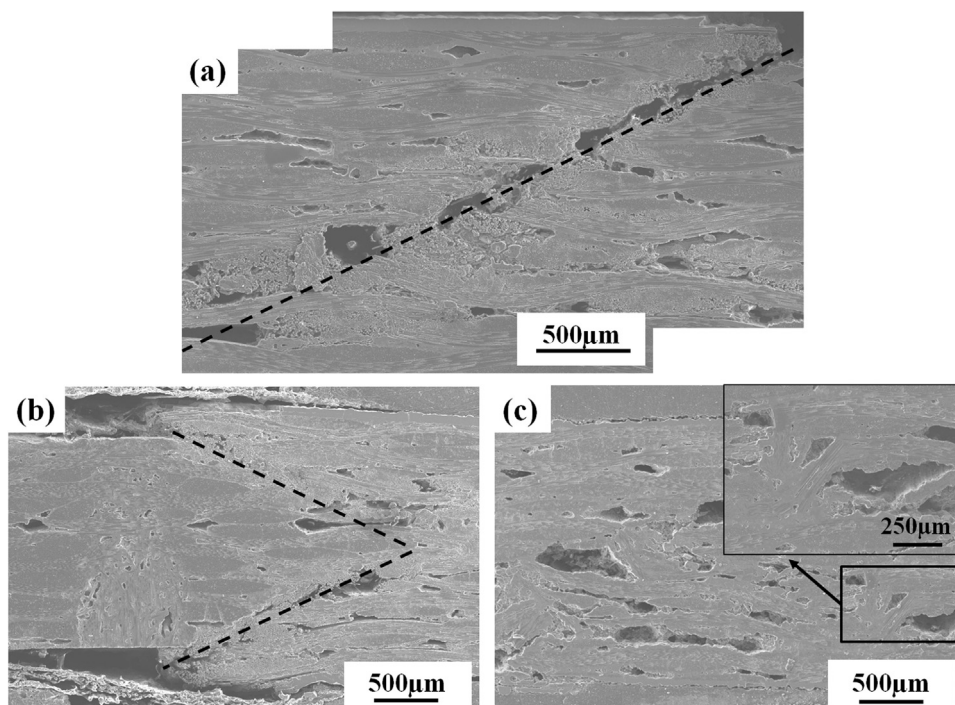


Fig. 6. SEM micrograph of a polished surface for the failed compression specimens. (a) 1200 °C, (b) 1400 °C, and (c) 1600 °C.

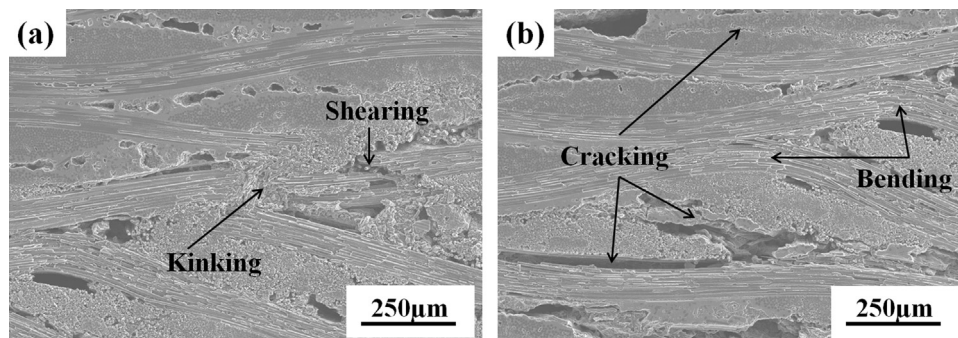


Fig. 7. SEM images of polished surfaces for the failed compression specimens at (a) 1400 °C and (b) 1600 °C.

cracking in 90° fiber bundles (Fig. 7(a)). At 1600 °C, 0° fiber bundles undergo bending, as well as the debonding of 0°/90° fiber bundle interface (Fig. 7(b)).

3.3. Damage accumulation

Above results show that the compressive damage modes of 2D-C/SiC composites involve 0° fiber bundle fracture, matrix cracking, interbundle cracking in 90° fiber bundles, and debonding between 0°/90° fiber bundles. A loading/unloading test was carried out to further determine damage evolution during the compression. Fig. 8 shows a SEM image of the 2D-C/SiC unloaded from 100 MPa. The composite is almost intact, as indicated by no crack and debonding in the composite. Only the shape and size of some original pores in the material changed. As shown in Fig. 8, the pore between 90° fiber bundles A and B closed, because the two fiber bundles approached and touched together. Moreover, fiber bundle A moved towards fiber bundle B, resulting in the opening of pore C and smaller dimension of pore D located between fiber bundle A and the adjacent 0° and 90° fiber bundles. The variation in the dimension of these intrinsic pores indicates that the material only suffered from the microbuckling of 0° fibers under 100 MPa.

Fig. 9 shows further damage evolution when the sample was re-loaded to 200 MPa and unloaded three times. Fig. 9(a) shows the as-received microstructure of 2D-C/SiC. Large intrabundle pores between fibers bundles and small interbundle pores between 90° fiber bundles are observed. The first loading to 200 MPa results in the closing of some interbundle pores and large intrabundle pores, as shown in Fig. 9(b). After the second loading to 200 MPa, the intrabundle pore reopened, accompanied by matrix cracking between the intrabundles. Moreover, a significant amount of interbundle pores were generated (Fig. 9(c)). Moreover, a longitudinal crack with a length of about 84 µm was initiated in the 90° fiber bundles. After the specimen was loaded to 200 MPa three times, the longitudinal crack underwent substantial growth and almost penetrated the 90° fiber bundle. The length of crack increased to about 470 µm. Moreover, the intrabundle pores connected and became a larger pore.

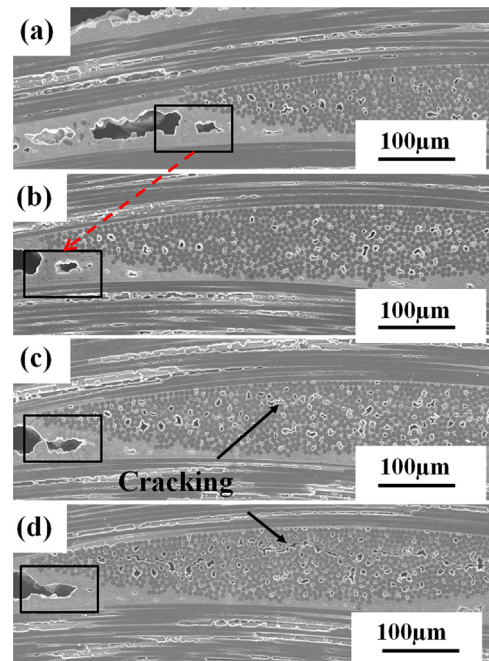


Fig. 9. Damage evolution during the repeated loading of 200 MPa. (a) 0 MPa, (b) first time 200 MPa, (c) second time 200 MPa, and (d) third time 200 MPa.

A further increase in the loading to 300 MPa resulted in rupture of the specimen. Fig. 10 shows the SEM image of the broken specimen. Area B in Fig. 10(b) corresponds to area A of Fig. 10(a). To clearly identify damage accumulation, the microstructure of as-received specimen is also shown, where the location of cracking is indicated by dashed lines. Two types of cracks were observed in the specimen. Type one belongs to a longitudinal crack distributed in 90° fiber bundle. One or two longitudinal cracks were observed in some 90° fiber bundles. This type of crack contributes to the separation of 90° fiber bundles or

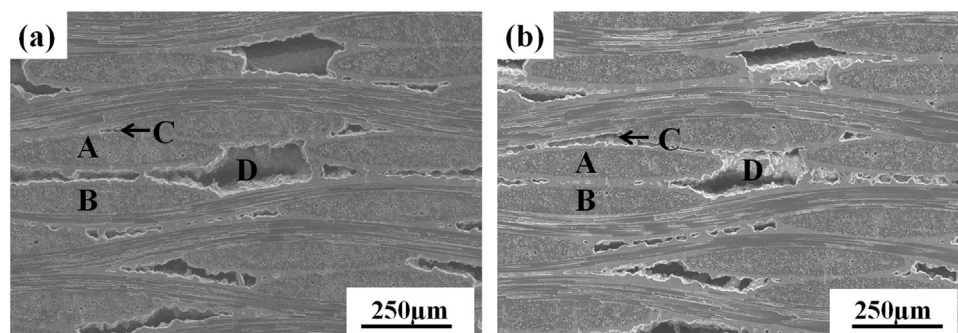


Fig. 8. In-situ observation of 2D-C/SiC composite (a) before and (b) after a loading of 100 MPa.

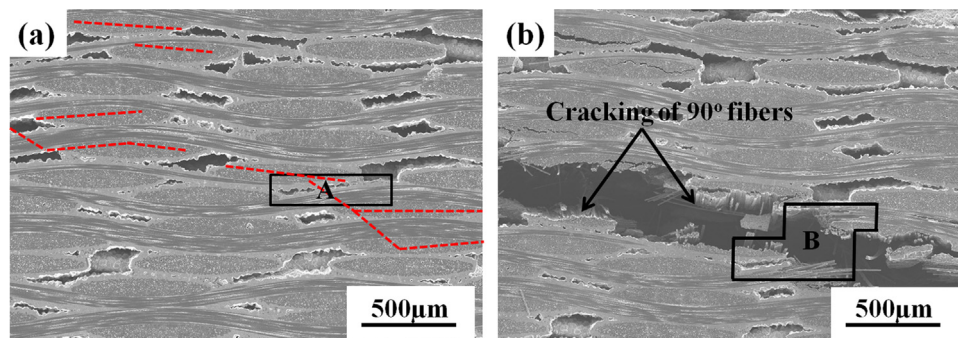


Fig. 10. Damage zones after the loading/unloading test., Microstructure of the specimen is shown in (a) and the dashed lines indicate the damage zone in (b).

delamination in local areas and cannot cause the failure of composite. Type two crack lies in the matrix and 0° fiber bundles and inclined to the loading direction by about 20° . This type of crack will contribute to the failure of 0° fiber bundles.

Based on the results, it can be concluded that the compressive damage evolves from the opening and closing of original pores, and followed as the sequence of formation and growth of longitudinal interbundle cracks, separation of 90° fiber bundles by longitudinal cracks, matrix cracking initiated from intrabundle pores, propagation of matrix cracks into 0° fiber bundles, and connection of the crack in 0° fiber bundles and the longitudinal cracks in 90° fiber bundles. The damage accumulation eventually leads to the complete failure of composite.

3.4. Discussion

Fig. 11 shows a schematic damage evolution of 2D-C/SiC loaded in compression. Under a lower stress, the buckling of 0° fiber bundles occurs, leading to the opening and closing of intrinsic pores, as shown in Fig. 11(b). Further increase in stress will result in propagation of original cracks and generate new microcracks in 90° fiber bundles, as shown in Fig. 11(c). The new microcracks in 90° fiber bundles are parallel to the loading direction, and the orientation of cracks is caused by further buckling of 0° bundles, causing the debonding of $0^\circ/90^\circ$ fiber bundle interface and longitudinal cracks in 90° fiber bundles. This is the reason for the macroscopic delamination of 2D-C/SiC composite. These cracks continue to grow to 0° fiber bundles at a higher stress and cause the failure of 0° fiber bundles, as shown in Fig. 11(d). The damage of 0° fiber bundle includes fiber kink, fiber bending, and shear fracture [13–15].

The above damage evolution shows that the compressive damage of 2D-C/SiC composite starts at the buckling of 0° fiber [16,17]. 0° fiber bundles are supported by SiC matrix during the compression, as a result, the compressive strain in SiC matrix will lead to the buckling of 0°

fiber. Therefore, the modulus of SiC matrix plays an important role in the buckling of 0° fiber. In other words, the compressive damage of 2D-C/SiC starts from the compressive strain in SiC matrix. At 1600°C , the matrix atoms undergo diffusion-oriented flow, resulting in a lower matrix modulus [18]. In a macroscopic view, the matrix appears to be plastic deformation. At the same time, the directional flow of SiC matrix atoms shortens the local fiber support and causes severe damage. Therefore, the composite undergoes plastic deformation at 1600°C , and the compressive strength decreases.

4. Conclusions

- (1) The compressive strength of 2D-C/SiC composites gradually decreases with the increase in temperature, and the materials mainly show shear and delamination when the samples were tested at $1200\text{--}1600^\circ\text{C}$.
- (2) Two types of cracks were observed on the macroscopic section. The inclined cracks indicate that the material was fractured by shearing, and the parallel cracks indicate delamination between the different layers of material fiber braid.
- (3) The damage of 2D-C/SiC composites is an accumulation process. The compressive damage of 2D-C/SiC starts at the buckling of 0° fiber and evolves as follows: opening and closing of original pores, formation and growth of longitudinal interbundle cracks, separation of 90° fiber bundles by longitudinal cracks, matrix cracking initiated from intrabundle pores, propagation of matrix cracks into 0° fiber bundles, and connection of the crack in 0° fiber bundles and the longitudinal cracks in 90° fiber bundles.
- (4) The modulus of matrix of 2D-C/SiC decreases at 1600°C and undergo plastic deformation under compressive loading. The decrease in matrix modulus and loss of stability of matrix decreases the compressive strength.

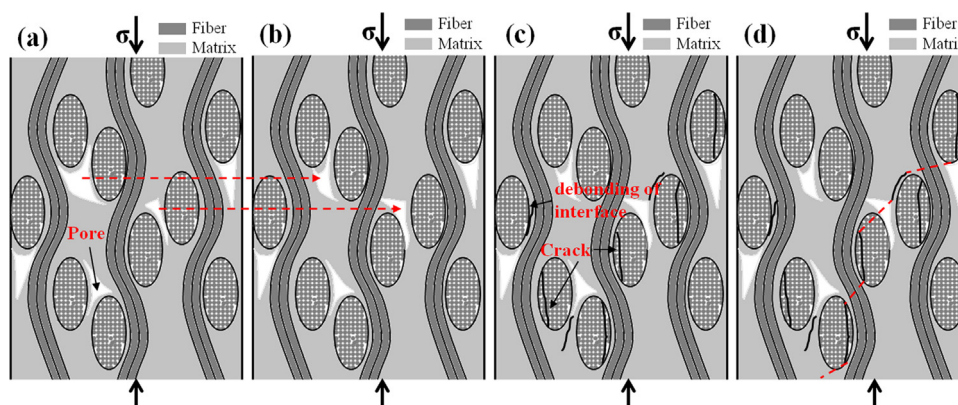


Fig. 11. Schematic diagrams of compressive damage evolution for 2D-C/SiC composite. (a) original, (b) opening and closing of intrinsic pores, (c) microcracks in 90° bundles, and (d) failure of 0° bundles.

Acknowledgements

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